

# MACHINE LEARNING COMPETITION REPORT — QUANTITATIVE EQUITY PORTFOLIO GENERATION: A SUB-ENSEMBLE LEARNING-TO-RANK APPROACH

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**ABSTRACT.** This paper details the development of a fully automated, machine-learning-driven quantitative equity portfolio targeting the CSI500 universe. By reframing the traditional return prediction problem into a Learning-to-Rank paradigm, we developed a Domain-Specific Sub-Ensemble utilizing Gradient Boosted Decision Trees (XGBRanker). We emphasize robust feature engineering via cross-sectional ranking and strict prevention of data leakage through chronological embargoes. Furthermore, we introduce an Exponential Conviction Allocator for portfolio construction. Evaluated strictly out-of-sample against a highly tuned official baseline, our methodology achieved an excess return of +173.48 Basis Points (BPS) over a 5-day forward horizon.

## 1. INTRODUCTION

The formulation of algorithmic trading strategies in modern quantitative finance relies heavily on extracting signal from noisy, non-stationary market data. The primary objective of this project was to design an end-to-end machine learning pipeline capable of constructing a compliant, high-alpha equity portfolio (minimum 30 stocks, maximum 10% weight per asset) from a provided universe of CSI500 features. To meet the stringent evaluation criteria, this paper is structured into four primary components: Factors, Models, Results, and Analysis.

## 2. FACTORS: SIGNALS AND FEATURE ENGINEERING

Financial time-series data is notoriously susceptible to macro-economic shocks, volatility clustering, and regime shifts. To extract stable alpha, we transformed the raw numeric features provided in the dataset prior to model ingestion.

**2.1. Cross-Sectional Ranking.** Rather than utilizing raw price, volume, or momentum values, we applied daily cross-sectional ranking to all features. For a given feature  $f$  on day  $t$ , each stock's value was converted into a percentile rank  $R_{i,t} \in (0, 1]$ . This normalization achieved strict market-neutrality; if a macro shock causes the entire market to decline, a high-momentum stock retains its high relative rank, preserving the predictive signal while filtering out systematic noise.

**2.2. Domain Feature Grouping.** To prevent "feature drowning"—a phenomenon where dominant features (e.g., raw volatility) overshadow subtle but predictive signals (e.g., technical indicators)—we manually grouped the dataset into three logical domains:

- **Momentum Features:** Capturing price trends, rate of change, and historical returns.
- **Volume Features:** Capturing liquidity, turnover rate, and trading amount.
- **Technical Features:** Serving as a catch-all for remaining distinct indicators.

This deliberate grouping formed the foundation for our model architecture.

### 3. MODELS: ARCHITECTURE AND PORTFOLIO CONSTRUCTION

Our pipeline departs from traditional regression models by adopting a Learning-to-Rank (LTR) framework, ensuring the model focuses exclusively on the relative ordering of assets rather than absolute return magnitudes.

**3.1. The Learning-to-Rank Paradigm.** We utilized the `rank:pairwise` objective within the XGBoost framework. To satisfy the discrete relevance requirements of LTR algorithms, we implemented Target Integer Bucketing, converting continuous forward returns into daily 5-level quantiles (0 to 4, where 4 represents the highest outperformance).

**3.2. Domain-Specific Sub-Ensembles.** Instead of a monolithic model, we trained three independent Gradient Boosted Decision Tree (GBDT) experts corresponding to our feature domains (Momentum Expert, Volume Expert, Tech Expert). Upon inference, each expert generates a score for every stock. These scores are cross-sectionally ranked, and the final stock prediction is the arithmetic mean of the experts' ranks. This ensemble drastically increases model diversity and stability.

**3.3. Strict Validation and Leakage Prevention.** We enforced a strict chronological split (`train`, `val`, `test`). Crucially, we implemented an **Embargo Period** (equal to the forward horizon) between all splits to guarantee zero look-ahead data leakage. We utilized the validation set to monitor the Normalized Discounted Cumulative Gain (NDCG) for early stopping, preventing overfitting. Our hyperparameters, optimized for safety and generalization, are centralized for 100% reproducibility (Table 1).

**3.4. Exponential Conviction Allocator.** To translate ranked predictions into portfolio weights, we designed a custom allocator. Stocks are sorted descending by ensemble score. The weight  $w_r$  for a stock at rank  $r$  is initialized using an exponential decay function:

$$w_r \propto \exp(-\lambda \cdot r) \tag{1}$$

where  $\lambda$  is our configured decay factor. The weights are then strictly bounded to a maximum of 10%, with excess capital redistributed proportionally to lower-ranked

| Hyperparameter                                     | Value                      |
|----------------------------------------------------|----------------------------|
| Number of Estimators ( <code>n_estimators</code> ) | 1500                       |
| Learning Rate ( <code>learning_rate</code> )       | 0.03                       |
| Max Depth ( <code>max_depth</code> )               | 4                          |
| Min Child Weight                                   | 10                         |
| Subsample / Colsample by Tree                      | 0.8 / 0.8                  |
| Objective                                          | <code>rank:pairwise</code> |
| Decay Factor ( $\lambda$ )                         | 0.001                      |

TABLE 1. Model and Allocator Configuration (CONFIG)

assets, guaranteeing both high-conviction tilts and regulatory compliance (minimum 30 assets).

#### 4. RESULTS: OUT-OF-SAMPLE PERFORMANCE

To validate the strategy, we simulated a live trading environment using our locked, unseen `test` set. Predictions were generated using feature data exactly one day prior to the start of the 5-day evaluation window ("As-Of" date logic).

When scored via the official evaluation metric against the repository's highly tuned baseline model, our methodology yielded an excess return of **+173.48 Basis Points (BPS)** relative to the baseline.

For final submission inference, we retrieved the absolute latest market data. To preserve the optimal stopping points discovered via the validation NDCG and avoid overfitting to recent noise, we explicitly chose *not* to refit the models on the full dataset, relying purely on the generalized trees.

#### 5. ANALYSIS: INSIGHTS AND ITERATIONS

Throughout the development of this pipeline, several iterations revealed the underlying mechanics of quantitative modeling.

##### 5.1. What Worked and Why.

- **Sub-Feature Ensembles:** This was the primary driver of alpha. By forcing independent models to learn from isolated feature groups, the ensemble captured orthogonal signals that a monolithic model simply ignored in favor of dominant volatility metrics.
- **Cross-Sectional Feature Ranking:** This transformation effectively standardized different market regimes, allowing the decision trees to identify consistent, stationary splitting rules.
- **Optimal Decay Rate ( $\lambda = 0.001$ ):** We discovered that this specific decay rate provided a mathematically balanced basket akin to an equal-weight portfolio, but with a slight, safe tilt toward our absolute highest-conviction ideas.

## 5.2. What Did Not Work and Why.

- **Single Monolithic GBDT:** A standard XGBoost model trained on all features merely tied the baseline. It lacked the capacity to generalize across noisy financial datasets without severe overfitting.
- **Simple Bagging Ensemble:** Averaging multiple GBDTs trained on the identical full dataset fell short of the baseline. Without feature subsetting, the trees were highly correlated, rendering the ensemble redundant.
- **High Decay Rate ( $\lambda = 0.15$ ):** An aggressive allocation curve failed catastrophically. The math forced exactly 10% of capital into our top 10 stocks and 0% to the remaining 40, resulting in immense single-stock concentration risk that destroyed alpha during random macro events.

Ultimately, our outperformance underscores the Pareto Principle in quantitative finance: massive alpha was achieved not through brute-force computation or hyper-stacking complex models, but through elegant data engineering, strict leakage prevention, and mathematically sound portfolio allocation.

## 6. CONCLUSION

This study demonstrates that a structured Learning-to-Rank approach, combined with Sub-Feature Ensembles and cross-sectional normalization, yields highly robust out-of-sample alpha. By prioritizing mathematical stationarity and risk-conscious allocation constraints, the resulting pipeline is both highly performant and deeply resilient to market noise.

## CODE AVAILABILITY

The full pipeline, including data preprocessing, model training, and the final submission generation script, is available at the following repository:

<https://github.com/JustinJiBian/Machine-Learning-Trading-Competition-NYUSH-SP26/tree/main>

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